RW–10053 **541402/545402**

M.C.A. DEGREE EXAMINATION, APRIL 2015

Computer Applications

DISTRIBUTED COMPUTING

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define: middleware.

2. List out the threads to distributed systems.

3. Define: marshalling.

4. What is RMI?

5. Draw the layers of operating system.

6. List out the distributed file system requirements.

7. Define: coordinated universal time.

8. What are naming services?

9. Define: deadlock.

10. What are video on demand services?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain about resource sharing and web.

Or

(b) Explain the security model in detail.

12. (a) Write note on group communication.

Or

(b) Explain the Remote procedure call.

13. (a) Explain the threads in distributed systems.

Or

(b) Explain the file service architecture in detail.

14. (a) Explain the network time protocol.

Or

(b) Write note on distributed mutual exclusion.

15. (a) Explain the nested transaction in detail.

Or

(b) Write note on transaction recovery.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the Internet protocols in detail.

17. Describe the communication between distributed objects.

18. Explain the various security techniques.

19. Describe domain name system.

20. Describe concurrency control in distributed transactions.

RW–10052 **541401/545401**

M.C.A. DEGREE EXAMINATION, APRIL 2015

Computer Applications

VISUAL PROGRAMMING

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List the properties of Command Button.

2. What are the data types in VB?

3. Write a note on: Abs( ) function.

4. Write a note on: The Line Control.

5. Distinguish between objects and classes.

6. What is MFC?

7. What are Mouse Events?

8. Define: Serialization.

9. What is meant by ODBC?

10. What are user interface classes? List them.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Discuss the various parts in the Opening Screen of VB Environment.

Or

(b) Discuss the usage of Labels and Text Boxes in VB with examples.

12. (a) How to use Grid Controls in VB? Explain with examples.

Or

(b) Write Short Notes on: File System Controls in VB.

13. (a) Discuss about various VC++ Components.

Or

(b) Write about Document View Architecture.

14. (a) Explain the usage of various Keyboard Events with examples.

Or

(b) Write about Splitter Windows in VC++ with examples.

15. (a) How to create User Defined DLLs? Explain briefly.

Or

(b) Write Short Notes on: OLE.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Develop a VB GUI based environment for payroll processing system.

17. Explain any five Built-In functions and User-Defined functions in VB with suitable examples.

18. Describe Resources and Dialog Boxes in VC++ with examples.

19. Explain how to develop Dialog Based Applications in VC++.

20. Explain about communicating with other applications with suitable examples.

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RW–10027 **541202/545202**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Applications

OPERATING SYSTEMS

(Common to M.C.A. (R)/W/E.)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the objectives of OS.

2. What is the purpose of system call?

3. Give the reasons to suspend a process.

4. What do you mean b throughput?

5. What operations can be performed on a semaphore?

6. What do you mean by safe state?

7. What is fragmentation?

8. What are the criteria to be followed while choosing of file organization?

9. What do you mean by file descriptor?

10. List the disk performance parameters.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the different types of I/O methods.

Or

(b) Describe the different categories of system programs.

12. (a) Explain the structure of process control block.

Or

(b) Explain the criteria for CPU scheduling.

13. (a) Describe the dining-philosophers problem.

Or

(b) Explain the usage of resource-allocation graph.

14. (a) Explain the concept of swapping.

Or

(b) How protection can be provided for files?

15. (a) Explain how free space is maintained using Bit vector.

Or

(b) Write about bad blocks.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. With an example, explain how system calls are used? Also describe the different types of system calls.

17. Explain the Round-Robin scheduling algorithm with an example.

18. Explain the deadlock prevention methods.

19. Describe the paging system in detail.

20. What are the disk scheduling algorithms available and explain any two of them?

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RW–10028 **541203/545203**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Science

DESIGN AND ANALYSIS OF ALGORITHMS

(Common for MCA(R)/MCA (W/E))

(CBCS – 2012 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Write the essential properties and the performance measures of an algorithm.

2. What is big O notation? Arrange the following functions by growth value :

   $N, N^2 N\log_2 N, N\log_2^2 N, N^3, 2^N$.

3. What is the importance of sorting?

4. Explain the difference of between bubble sort quick sort.

5. State Dynamic programming.

6. Write a short note on Knapsack problem.

7. Define a Binary heap.

8. Define Graph.

9. State Back Tracking.

10. Define Spanning Tree.

Part B $(5 \times 5 = 25)$

Write **all** the questions.

11. (a) Describe the analysis of Algorithms.

Or

(b) Explain briefly the Disjoint Set Operations.

12. (a) Write a short note on Binary Search.

Or

(b) Write an algorithm for computing Fibonacci numbers.

13. (a) Describe the job sequencing problem.

Or

(b) Explain briefly the Greedy Technique.

14. (a) Explain the Breadth First Search.

Or

(b) Explain the procedure for Insertion Sort.

15. (a) Explain Branch and Bound Method.

Or

(b) Describe travelling Salesman Problem.

Part C $(3 \times 10 = 30)$

Write any **three** questions.

16. Discuss in detail the Asymptotic Notations.

17. Discuss the basic strategy and algorithm of Quicksort.

18. Explain in detail the Kruskal's Algorithm.

19. Explain the procedure to perform Heapsort in detail.

20. Explain in detail the Graph Traversals.

RW–10029 **551203/541204/545204**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

COMMUNICATION SKILLS

(Common for M.C.A. (R)/M.C.A. (W/E)/M.Sc., Comp.Sci.)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What does downward communication mean?

2. Define intrapersonal communication

3. How would you thank a person?

4. What is the mode of greeting a person?

5. Define proxemics.

6. What is the purpose of using models?

7. What does employment interview mean?

8. What is the role of the chairperson in a meeting?

9. Define memo.

10. What is an agenda?

Part B (5 × 5 = 25)

Answer each of the following questions, choosing either (a) or (b).

11. (a) Analyse the different levels of communication.

Or

(b) Bring out the barriers to communication.

12. (a) Discuss the importance and attributes of soft skills.

Or

(b) Bring out the do's and don'ts of telephone conversation.

13. (a) Analyse the purpose of audio visual aids.

Or

(b) How would you prepare for a presentation?

14. (a) How are meetings organised?

Or

(b) What are the skills essential for team work?

15. (a) Analyse the purpose of preparing minutes.

Or

(b) Discuss the format of writing a report.

Part C (3 × 10 = 30)

Answer any **three** of the following questions.

16. Examine the importance of communication

17. Discuss the nuances in the modes of conversation.

18. Analyse the mode of planning in presentation skills.

19. What are the advantages of group discussion?

20. Draft a job application letter enclosing your curriculum vitae for the post of Software Engineer.

RW–10030 **541205/545205**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

ACCOUNTING AND FINANCIAL MANAGEMENT

(Common For M.C.A (R)/M.C.A. (W/E))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What is the need for accounting concepts?

2. Differentiate between fund and cash.

3. State the important objectives of cost accounting.

4. State the objectives of budgetary control.

5. What do you understand by time value of money?

6. What is risk-return trade-off?

7. Differentiate between an equity share and a preference share.

8. What is stock dividend?

9. What is Zero Base Budgeting(ZBB)?

10. State the objectives of standard costing.

Section B

$(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) What do you mean by Financial Management? Explain its basic aspects.

Or

(b) Explain the importance of Break-even Analysis in decision making.

12. (a) Explain the managerial applications of Marginal costing.

Or

(b) Explain the concept of cost of capital and its significance while taking financial decision.

13. (a) Distinguish between budgetary control and standard costing.

Or

(b) What are the implications of a stable dividend policy?

14. (a) XYZ Ltd. is considering purchase of a machine costing Rs. 1,20,000. The machine will have a life of 5 years and no scrap value. The cash inflows from the machine are as follows :

Year	Cash inflow (Rs.)
1	30,000
2	45,000
3	50,000
4	55,000

Calculate the pay back period of this machine.

Or

(b) From the following, compute the Break Even Point in units and Break Even Point in value. Also compute the profit at 90% capacity when the total capacity is 500 units.

	Rs.
Variable cost of the product	200
Selling price per unit	250
Total fixed cost	12,000

15. (a) The following information relates to a flexible budget at 60% activity. Prepare a budget for production at 80%.

Production at 60% capacity 600 units

Materials	Rs. 100 per unit
Labour	Rs. 40 per unit
Direct Expenses	Rs.10 per unit
Factory Overheads	Rs. 40,000 (40%fixed)
Administration Expenses	Rs. 30,000 (60%fixed)

Or

(b) A company has the following capital structure. Calculate Weighted Average Cost of Capital.

Securities	Book value in Rs.	After – tax cost
Equity shares	9,00,000	15%
Retained earnings	3,00,000	15%
Preference share capital	2,00,000	8%
Debentures	16,00,000	6%

Section C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Outline the methods and tools of financial management.

17. What is Capital Budgeting? What are the various factors influencing Capital Budgeting decision?

18. Discuss the factors governing determinants and constraints of Dividend Policy.

19. (a) At the time of his retirement, R is offered the following options, by his company.
    (i) Receive Rs. 60,000 immediately,
    (ii) Receive Rs. 20,000 immediately, Rs.30,000 at the end of 4 years and Rs. 30,000 at the end of 8 years.
    (iii) Receive Rs. 10,000 now and another 9 annual payments of Rs. 10000 each. If the rate of interest expected by R is 10%, which option should he accept?

    (b) A deposits annually Rs. 4,000 in a bank which offers 10% interest per annum. What will be the value of the deposit at the end of 5 years, assuming
    (i) The first deposit is made immediately.
    (ii) The first deposit is made at the end of one year from now.

20. From the following capital structure of a company, compute the overall cost of capital using Book value weights

Particulars	Book value (Rs.)
Equity share capital (Rs. 10 per share)	45,000
Retained earnings	15,000
Preference share capital	10,000
Debentures	30,000

The after tax cost of different sources of finance is as follows. Equity share capital 14%. Retained earnings 13%, preference share capital 10% and Debentures 5%.

RW–10031

541401/ 545401

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015.

Computer Science

VISUAL PROGRAMMING WITH .NET

(Common for M.C.A. (R)/M.C.A. (W/E))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the role of Tool box in windows programming?

2. What is .NET?

3. What is bookmark?

4. Define Enums.

5. Define Icon.

6. What do you mean by delegates?

7. How the windows plays an important role in visual programming?

8. What is the root namespace of the .NET Framework?

9. What is ASP.NET?

10. What is MVC?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write down the benefits of NET.

Or

(b) How will you create and display the window?

12. (a) What is class? How are they defined in Visual Programming Languages?

Or

(b) Write a note on Visual Studio Code Editor.

13. (a) Describe in detail the delegates and events in .NET.

Or

(b) How do array and generics play a predominant role in creating .NET applications?

14. (a) Discuss in detail the following :

(i) Call stack Window

(ii) Quick watch Window.

Or

(b) What is database? How is it useful in generating records in .NET?

15. (a) Bring out the importance of MVC architecture in developing web applications.

Or

(b) What is Lay out? List out any two Panel Layout with an example.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the basis of .NET Framework and features of .NET in detail.

17. Discuss inheritance concept in detailed with suitable example.

18. What is class designer? How the code is generated using class designer? Discuss in detail with an example.

19. What is a database? Explain the steps to be followed to create a database using the Data Manager.

20. Explain the basis of ASP.NET application configuration procedures in steps with real world example.

RW–10032

541402/ 545402

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015.

Computer Science

DISTRIBUTED COMPUTING

(Common for M.C.A. (R)/WE)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are the advantages of Distributed systems over independent Pcs?

2. What are the challenges of Distributed computing?

3. What do you mean by Interprocess communication?

4. What is Remote Procedure Call?

5. Define Process.

6. What is meant by symmetric cryptosystem?

7. What is Domain Name System?

8. What is Logical Clock?

9. What is the role of Locks in transaction?

10. What are the conditions for distributed deadlock?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the fundamental models of distributed system.

Or

(b) Explain the layers of Internet protocol.

12. (a) Explain the Client-server communication model.

Or

(b) Explain the Java RMI with an example.

13. (a) Explain the Threads in distributed systems.

Or

(b) Explain the Andrew File system.

14. (a) Explain the X.500 Directory services.

Or

(b) Explain the purpose of distributed mutual exclusion.

15. (a) Explain the Time stamp ordering with an example.

Or

(b) Discuss transaction recovery.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the various types of networks.

17. Explain with an example, the communication methods between distributed objects.

18. Explain the Cryptographic algorithms.

19. Explain the Multicast Communication Related problems.

20. Explain the distributed Multimedia systems. Explain its elements.

RW–10033 **541403/545403**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Science

WEB TECHNOLOGY

(Common for MCA(R) /MCA (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Cascading?

2. What is DHTML?

3. What is Persistence?

4. Define Cookies.

5. What is Servelets?

6. Define HTTP Request.

7. What is MVC?

8. Define Conditional Processing.

9. What is Package?

10. What is JSP?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss briefly the XML Schema.

Or

(b) What is HTML? Write any four Common Tags in HTML.

12. (a) Mention the advantages of Java Beans.

Or

(b) Discuss about Introspection.

13. (a) List and Discuss any four HTTP request response methods.

Or

(b) Write short notes on :

(i) Cookies.

(ii) Session Tracking.

14. (a) Write a note on Error Handling and Debugging.

Or

(b) How Request and Response plays an important role in web services?

15. (a) How JSP is helpful in developing web pages?

Or

(b) Discuss the steps involved in deploying java beans in a JSP page.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the features and roles of XML as Choice for web services.

17. What is EJB? Explain the role of EJB in handling web services.

18. What is a session? Explain how client state is maintained in session and also explain session tracking and session management with an example.

19. Explain Tomcat application server and its uses.

20. How a Design an online book selling application using JSP or ASP. Consider a login validation page and also a search page for searching a book.

RW–10034 **541552/545552**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Science

SOFTWARE PROJECT MANAGEMENT

(Common for M.C.A. (R)/ M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a Project?

2. What is feasibility study?

3. Distinguish between software project and other project.

4. Define software Prototyping.

5. Define 'Performance' and 'predictive' measures.

6. What is work flow?

7. Define Critical path.

8. What is meant by Risk?

9. What are the activities to be scheduled for a project?

10. List out the attributes of a Successful leader.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Why Software Project Management is important?

Or

(b) Write a short note on Software Cost Estimation.

12. (a) Explain the overview of project planning in stepwise.

Or

(b) What is software management? Explain its activities.

13. (a) Describe briefly the work flow of the process.

Or

(b) In what way the periodic status assessment helps in developing an Organization.

14. (a) Write a note on Cost Monitoring.

Or

(b) What is the role and responsibilities of an Organization in handling projects?

15. (a) Write a detailed note on modem process transitions.

Or

(b) What are the various quality indicators included in software Project management?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With a neat sketch, explain the principle of Waterfall Model.

17. Describe briefly the different phases of Life cycle in handling software Project management.

18. What is model based software architecture? Distinguish the Management Perspective and Technical perspective in project management.

19. How planning is helpful in estimating the needs of an Organization?

20. What are the Seven Core Metrics followed in controlling the project for the betterment of an Organization?

RW–10035

541561/ 545561

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Science

SOFT COMPUTING

(Common for M.C.A. (R)/M.C.A. (W/E))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by Soft computing?

2. What are the applications of Soft computing?

3. What is a Boltzman machine?

4. What is ART Network?

5. Define Fuzzy set. Give example.

6. What is meant by Fuzzification?

7. Write any Two relationships in Fuzzy.

8. What are the applications of Fuzzy logic?

9. What are the steps involved in Genetic algorithm?

10. What are the applications of Genetic algorithm?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the Neural Network architecture.

Or

(b) Explain Pitts model.

12. (a) Explain Perceptron Network with diagram.

Or

(b) Explain Hopfield Network with an example.

13. (a) Explain the operations on Fuzzy set.

Or

(b) Explain the methods Membership value assignments.

14. (a) Explain Fuzzy reasoning with example.

Or

(b) Explain the Formation of rules.

15. (a) Compare Traditional and Genetic algorithm.

Or

(b) Explain the elements of Genetic algorithm.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the Hebb Network.

17. Describe any two Unsupervised learning network.

18. Explain Fuzzy equivalence and Tolerance relation with an example.

19. Explain :

    (a) Fuzzy Decision making

    (b) Fuzzy Logic control systems.

20. Explain the Genetic operators with example.

RW–10036 **541101/545101**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Digital Computer Organization Computer Application

DIGITAL COMPUTER ORGANIZATION

(Common for MCA(R)/MCA(W/E))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define 2's complement of a number.

2. Simplify $y = A\overline{B} + (\overline{A} + B)C$.

3. Write about floating point representation of a number with example.

4. Define Error Detection Code.

5. Write the types of computer instructions.

6. What are the different types of Addressing modes?

7. Give the definition for data transfer and data manipulation.

8. What is relative addressing?

9. Give some examples for peripheral devices.

10. Differentiate between main memory and Auxillary memory.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) Convert the following octal number to hexadecimal.

(i) $(4163)_8$

(ii) $(3643)_8$.

Or

(b) Explain don't care combination.

12. (a) Explain about 1–4 Demultiplexers.

Or

(b) Write a note on Gray code.

13. (a) Explain briefly about Timing and control.

Or

(b) Describe the data manipulation instruction.

14. (a) Describe the concept of stack organization with an example.

Or

(b) Write about instruction formats and number of addresses.

15. (a) Describe about serial communication.

Or

(b) Give a short notes on memory hierarchy.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Plot the logical Expression.

$ABCD + A\overline{B}\overline{C}\overline{D} + A\overline{B}C + AB$ on a 4–variable K–map. Obtain the simplified expression from the map

17. Explain the concept of counters in detail.

18. Explain about Instruction cycle organization of control registers.

19. Describe the different kinds of Addressing modes.

20. Explain the virtual memory concept in memory management.

RW–10037

541102/ 545102

M.C.A. DEGREE EXAMINATION, APRIL 2015

Computer Applications

C AND DATA STRUCTURES

(CBCS – 2012 onwards)

(Common for M.C.A. (R)/W.E.)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Lists the different data types of constants available in C.

2. Explain briefly the Scanf( ) and Printf( ) functions.

3. What do you mean by recursion?

4. How do you assign an address to an element of a pointer array?

5. What is tree data structure?

6. What are different ways of traversing a tree?

7. What is Garbage Collection?

8. Why tree is called a nonlinear data structure.

9. What is the maximum possible number of nodes in a binary tree at level 6?

10. What is the complexity of insertion sort in the worst case?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Discuss the decision making structure in detail.

Or

(b) Describe in detail formatted and unformatted functions in C.

12. (a) What is a linear array? Explain how two dimensional arrays are represented in memory.

Or

(b) Explain array of pointers.

13. (a) What is stock? Write the procedure for inserting and deleting an element from stack.

Or

(b) Explain the singly linked list and its operation.

14. (a) Explain with examples, the in-order, pre-order and post-order traversals of a binary tree.

Or

(b) Briefly explain the Hashing Techniques.

15. (a) Explain the bubble sort algorithm with an example.

Or

(b) What is selection sort? Sort the following array using selection sort method.

24 56 47 35 10 90 82 31.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss on passing arguments to a function with an example program.

17. With examples, explain dynamic memory allocation and deallocation methods.

18. Explain Insertion and Deletion in doubly linked list by giving the algorithm.

19. Construct the binary tree for the following :

    (a) In order : 3 5 6 8 12 15 18 19

    Pre order : A E F D J H I G B C

    Post order : 3 8 5 15 19 18 12

    (b) Pre order : 12 5 3 6 8 18 15 19

    In order : F E A H J I D B G C

20. Distinguish between Linear search and binary search methods. What are the best and worst cases for quick sort?

RW–10038

541103/ 545103

M.C.A. (WEP/R) DEGREE EXAMINATION, APRIL 2015

Computer Applications

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(CBCS – 2012 onwards)

(Common for M.C.A. (R)/WE)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Schema?

2. Define Weak Entity.

3. What is Integrity Constraint?

4. Give the View Update Operations.

5. Specify the uses of Database normalization.

6. What are the various Normal forms available in Distributed database Systems?

7. What is concurrency management in database system?

8. What is Buffer Management in RDBMS?

9. What is Indexing?

10. What is B+ Trees?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the pitfalls of RDBMS? Explain.

Or

(b) What is transaction management? Explain with an example.

12. (a) Explain the method of creating a View using SQL.

Or

(b) What is lossless join? Explain with an example.

13. (a) Explain functional dependency with an example.

Or

(b) Explain disaster recovery with an example.

14. (a) Explain the use of serializability graph with an example.

Or

(b) What is deadlock in transactions? Explain.

15. (a) Write a note on Dynamic Index Structure.

Or

(b) What are the different ways of performing the index in RDBMS?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain the Architecture and applications of database systems.

17. Explain Boyce Codd Normal form with an example.

18. Explain database recovery with respect to centralized DBMS.

19. Explain locking schemes and multi version Concurrency control with an example.

20. Write a detailed note on Primary and Secondary indexes. How index plays an important role in RDBMS?

RW–10039

541104/ 545104

M.C.A (R)/WE DEGREE EXAMINATION, APRIL 2015

DISCRETE MATHEMATICS

(CBCS – 2012 onwards)

(Common for M.C.A.(R)/WE 2012 and 2008 Batch)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define a semigroup with an example.

2. Show that the equivalence :
$P \to (Q \to P) \Leftrightarrow \neg P \to (P \to Q)$.

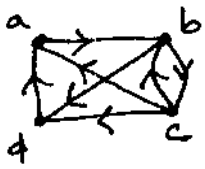
3. Verify whether the statement $(p \vee q) \to p$ is a tautology or not.

4. If A is a given finite set, $\rho(A)$ its power set and $\subseteq$ is the inclusion relation on the elements of $\rho(A)$, draw Hasse diagram of $(P(A), \subseteq)$ for $A = \{a, b, c\}$.

5. Show that $\neg Q, P \to Q \Rightarrow \neg P$.

6. Show that the set N of natural numbers is a semigroup under the operation $x * y = \max\{x, y\}$. Is it a monoid?

7. Write down the adjacency matrix for the graph.



8. Show that a graph G with n vertices, $n-1$ edges and no circuits is connected?

9. A pair of dice is tossed. Find the probability that one of the die is 2 if the sum is 6.

10. Let X_1 and X_2 be two random variables. Prove that $E(X_1 \pm X_2) = E(X_1) \pm E(X_2)$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$.

Or

(b) Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\neg R \vee P$ and Q.

12. (a) Let $X = \{1, 2, ...10\}$ and R be a relation on 'X' where $R = \{(x, y) : x + y = 10\}$. What are the properties of the relation R?

Or

(b) Show whether the following relations are transitive :

$R_1 = \{\langle 1,1 \rangle\}$ $R_2 = \{\langle 1,2 \rangle, \langle 2,2 \rangle\}$

$R_3 = \{\langle 1,2 \rangle, \langle 2,3 \rangle, \langle 1,3 \rangle, \langle 2,1 \rangle\}$.

13. (a) Let $S = N \times N$, N being the set a positive integers and *be an operation on S given by $(a,b) * (c,d) = (a+c, b+d)$. Show that S is a semigroup. Define $f : (s,*) \to (z,+)$ by $f(a,b) = a - b$ show that f is a homomorphism.

Or

(b) Show that the set of all elements a of a group of $\langle G,* \rangle$, such that $a * x = x * a$ for every $x \in G$ is a subgroup of G.

14. (a) Define the following terms. Give an example for each.

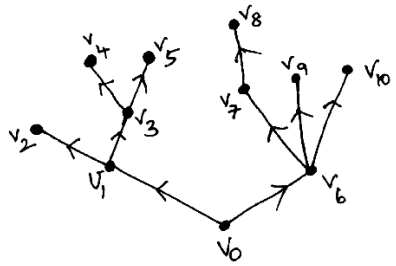
(i) Isomorphic graphs

(ii) Non isomorphic digraph

(iii) Circuit matrix.

Or

(b) Define binary tree. Obtain the binary corresponding to the tree.



15. (a) If A and B are independent events then $\overline{A}$ and $\overline{B}$ are also independent events. Also state addition theorem for probabilities.

Or

(b) Let X be a random variable with the following probability distribution.

x :	–2	–1	0	1	2	3
$p(X = x)$	0.1	k	0.2	2k	0.3	3k

Find the value of k, $P(X < 2), P(X \geq 2)$ and $P(-2 < X < 2)$.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. (a) Show the following using indirect method $S \to \neg Q, S \vee R, \neg R \rightleftarrows Q \Rightarrow \neg P$.

    (b) Show that the following premises are inconsistent. $P \to Q, P \to R, Q \to \neg R, P$.

17. Write all possible functions from $X = \{1,2\}$ to $Y = \{a,b,c\}$ and classify them into one-to-one, onto, neither one-to-one nor onto types of functions.

18. Prove that every finite group of order n is isomorphic to a permutation group of degree n.

19. (a) Define the following terms. Give an example for each graph.

    (i) Reflexive

    (ii) Transitive

    (iii) Symmetric

    (iv) Antisymmetric.

    (b) Prove that in a simple digraph, $G = (V, E)$, every node of the digraph lies in exactly one strong component.

20. State and prove Baye's theorem.

RW–10040 **551105/541105/545105**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Science

SOFTWARE ENGINEERING

(CBCS – 2012 onwards)

(Common for M.C.A. (R)/W/E and M.Sc. Computer Science)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What do you mean by Software Engineering?

2. Define Software myths.

3. Define feasibility study.

4. What is meant by data model?

5. List any two characteristics of object oriented design.

6. What do you mean by entity abstraction?

7. State the need for debugging.

8. Difference between alpha testing and beta testing.

9. What is risk?

10. Bring out the importance of software quality.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) Describe briefly about the waterfall model.

Or

(b) Write a note on unified process.

12. (a) State the fundamental principles of requirements analysis.

Or

(b) Explain the steps involved in object oriented modeling.

13. (a) List the different level of cohesion and explain each one.

Or

(b) What are the different types of architecture generally used for software development? Explain.

14. (a) Describe the black box testing.

Or

(b) Explain about the system testing.

15. (a) Discuss the Reactive vs. Proactive risk strategies.

Or

(b) Write notes on ISO 9000 quality standards.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain briefly about the incremental and spiral model.

17. Describe the various stages in requirements analysis.

18. Describe about the fundamental software design concept.

19. Discuss the white box testing methods in detail.

20. Write short notes on :

    (a) Objects and Object classes. (5)

    (b) RMMM Plan (5)

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RW–10041 **541301/545301**

M.C.A. DEGREE EXAMINATION, APRIL 2015

COMPUTER GRAPHICS

(Common for M.C.A. (R)/WE)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is frame buffer?
2. What is a work station?
3. What is rotation?
4. What is shearing?
5. Define Viewing transformation.
6. What is a Stroke device?
7. Define Clipping.
8. What do you mean by Composite transformation?
9. What do you mean by Isometric projection?
10. Define Depth Buffer.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the Random scan systems.

Or

(b) Write and explain the mid-point circle algorithm.

12. (a) Explain the homogeneous co-od system with example.

Or

(b) Explain Window to Viewport transformation.

13. (a) Explain the B-Spline curves.

Or

(b) Explain the Polygon surfaces.

14. (a) Discuss 3D display techniques.

Or

(b) Explain 3D viewing pipeline.

15. (a) Explain the concepts of 3D animation.

Or

(b) Discuss the Key frame systems.

Part C $(3 \times 10 = 30)$

Answer any **three** questions only.

16. Explain the Video display devices.

17. Write and explain Sutherland-Hodgman Polygon Clipping algorithm.

18. Explain the 3D basic illumination models.

19. Explain the 3D basic transformation with matrix and equation.

20. Explain any two visible surface detection methods.

RW–10042 **541303/545303**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Applications

COMPUTER NETWORKS

(Common for M.C.A. (R)/ M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Classify computer networks based on geographical distance.

2. Compare OSI and TCP\IP models.

3. What is MAC?

4. What do you know about stop and wait protocol?

5. What is selective flooding?

6. What is meant by traffic shaping?

7. Distinguish between Remote login and Remote procedure call.

8. What are the design issues of transport layer?

9. When would you use digital signature?

10. What is one time pads?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

All questions carry equal marks.

11. (a) Explain about twisted pair and fiber optics.

Or

(b) Explain the different types of topology.

12. (a) Write notes on error correcting codes and detecting codes.

Or

(b) Explain about :

(i) ALOHA

(ii) CSMA/CD.

13. (a) Explain any two dynamic routing algorithms.

Or

(b) Explain the hierarchical and multicast routing algorithms.

14. (a) Write about UDP.

Or

(b) Explain about SNMP.

15. (a) Explain the transposition and substitution ciphers method.

Or

(b) Describe the RSA public key algorithm.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Explain about OSI reference model.

17. Explain in detail about shading window protocol.

18. Explain about various congestion control algorithms.

19. In detail, explain about Transmission control protocol.

20. Explain the DES symmetric key cryptography algorithm.

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RW–10043 **541551/545551**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

First Semester

Computer Science

OBJECT ORIENTED ANALYSIS AND DESIGN

(Common for M.C.A. (R)/M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the advantages of Object Oriented System?

2. What is the purpose of static model?

3. What is meant by Super and Sub class?

4. What is meant by Association?

5. What do you mean by Class visibility?

6. What do you mean by Design pattern?

7. Define Database.

8. What do you mean by logical database?

9. What are the metrics used in testing?

10. Define Polymorphism testing.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the class diagram with suitable example.

Or

(b) Discuss the various relationships among objects.

12. (a) Discuss the Business process model.

Or

(b) Explain Aggregation with an example.

13. (a) Describe the documentation process.

Or

(b) Discuss Packages and Managing classes.

14. (a) Explain in detail about Client-Server computing.

Or

(b) Explain the View layer interface.

15. (a) Explain the class testing method with an example.

Or

(b) Explain User satisfaction test with example.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe Object Oriented Systems Development Life cycle.

17. Explain the different object modeling techniques.

18. Explain in detail the designing methods and protocols.

19. Describe Object Oriented Database Management System.

20. Explain Myer's Debugging principles.

RW–10044 **541560/545560**

M.C.A. DEGREE EXAMINATION, APRIL 2015

Third Semester

Computer Applications

RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2012 onwards)

(Common for MCA (R)/MCA (W/E)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the standard form of an LPP.

2. State the purpose of artificial variable.

3. State the fundamental properties of duality.

4. Define degenerate solution in transportation problem.

5. What is mixed IPP?

6. What is Gomory's constraint?

7. What is critical activity?

8. Give the formula to find the cost slope.

9. Classify the queueing system.

10. Define lead time.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Use graphical method to solve

Max $z = 2x_1 + x_2$

Subject to the constraints :

$x_1 + x_2 \leq 10$

$x_1 + x_2 \leq 6$

$x_1 - x_2 \leq 2$

$x_1 - 2x_2 \leq 1$

$x_1, x_2 \geq 0.$

Or

(b) Explain the method of penalty.

12. (a) Show that dual of dual is primal.

Or

(b) Consider the problem of assigning five operators to five machines. The assignment costs are given in the following table.

	M_1	M_2	M_3	M_4	M_5
A	7	7	–	4	8
B	9	6	4	5	6
C	11	5	7	–	5
D	9	4	8	9	4
E	8	7	9	11	3

Operator A cannot be assigned to machine M_3 and C cannot be assigned to machine M_4. Find the optimum assignment schedule.

13. (a) Find the optimum integer solution to the LPP:

Max $Z = x_1 - 2x_2$

Subject to the constraints :

$4x_1 + 2x_2 \leq 15$

$x_1, x_2 \geq 0$ and are integers

Or

(b) Describe the Branch-and-bound algorithm.

14. (a) Draw the network for the following:

B < E, F; C < G, L ; E, G < H;

L, H < I ; L < M; H, M < N;

H < J; I, J < P; P < Q.

Or

(b) Write about crashing.

15. (a) Describe the characteristics of queueing system.

Or

(b) A manufacture has to supply his customer with 600 units of his product /year. Shortages are not allowed and the storage cost amounts to Rs. 0.60/ unit/ year. The set up cost per run is Rs. 80. Find the optimum run size and the minimum average yearly cost.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Use two-phase simplex method to solve :

Min $Z = x_1 + x_2$

Subject to the constraints :

$2x_1 + x_2 \geq 4$

$x_1 + 7x_2 \geq 7$

$x_1, x_2 \geq 0.$

17. Use dual simplex method to solve :

Max $Z = -3x_4 - 2x_2$

Subject to the constraints :

$x_1 + x_2 \geq 1$

$x_1 + x_2 \leq 7$

$x_1 + 2x_2 \geq 10$

$x_2 \leq 3$

$x_1, x_2 \geq 0$

18. Find the optimum integer solution to the LPP

Max $Z = 2x_1 + 2x_2$

Subject to the constraints :

$5x_1 + 3x_2 \leq 8$

$x_1 + 2x_2 \leq 4$

$x_1, x_2 \geq 0$ and are integers

19. Draw the network and find the critical path for the following project:

A, C, D are start jobs

E > B, C ; F, G > D; H, I > E, F;

J > I, G; K > H; B > A.

Task :	A	B	C	D	E	F	G	H	I	J	K
a	4	5	8	2	4	6	8	5	3	5	6
b	8	10	12	7	10	15	16	9	7	11	13
m	5	7	11	3	7	9	12	6	3	8	9

20. Given an average arrival rate of 20 per hour, is it better for a customer to get service at a single channel with mean service rate of 22 customers or at one of two channels in parallel with mean service rate of 11 customers for each of the two channels. Assume that both queues are of M/M/C type.

RW–10046 **541201/545201**

M.C.A. (R)/WE DEGREE EXAMINATION, APRIL 2015

OBJECT ORIENTED PROGRAMMING AND C++

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define : Polymorphism.

2. Write the purpose of scope resolution operator with an example.

3. What are friend functions?

4. Write the purpose of 'this' pointer with an example.

5. What is meant by Operator Overloading?

6. What are Virtual Functions?

7. Define: Templates.

8. List down any four file mode parameters.

9. What is an Exception?

10. What should be placed inside a catch block?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write Short Notes on: Evolution of Programming Paradigm.

Or

(b) Discuss about Popular OOP Languages.

12. (a) What is Constructor Overloading? Discuss this with an example.

Or

(b) What are Self-referential Classes? Discuss with example.

13. (a) Write a C++ Program to illustrate Unary Operator Overloading.

Or

(b) Write about Pure Virtual Functions with examples.

14. (a) What are Class Templates? Explain with examples.

Or

(b) Discuss briefly on: Formatted Console I/O operations in C++ with examples.

15. (a) Write Short Notes on: Exception Handling Model.

Or

(b) What are Fault Tolerant Design Techniques? Discuss briefly about them.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the various elements of Object-oriented Programming with examples.

17. Explain different types of Constructors in C++ with example programs.

18. Explain Multilevel and Multiple Inheritances in C++ with example programs.

19. (a) Write a Function Template for finding the minimum value contained in an array.

    (b) Discuss briefly on: Error Handling Functions in C++ with examples.

20. Discuss the following with examples :

    (a) Exceptions in Constructors and Destructors.

    (b) Exceptions in Operator Overloaded Functions.

RW–10047 **541202/545202**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Application

OPERATING SYSTEM

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the important duties of OS?

2. Name the structure of OS.

3. What the term scheduling refers?

4. Write any two criteria for CPU scheduling.

5. What is meant by critical section?

6. What is semaphore?

7. When memory allocation problem arise?

8. What is meant by fragmentation?

9. What do you understand about disk scheduling?

10. Name any two types of directory structure.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Describe distributed system and the reason for its popularity.

Or

(b) Discuss about multiprocessor system and its draw back.

12. (a) Discuss process states with diagram.

Or

(b) Discuss process scheduling and CPU scheduling concepts.

13. (a) Describe the situations where deadlock may arise.

Or

(b) Explain Test-and-set synchronization method.

14. (a) Describe contiqous allocation methods.

Or

(b) Explain the fixed partition memory management scheme.

15. (a) Discuss the file concepts and file protection methods.

Or

(b) Explain Bit-vector free space management technique.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss:

    (a) Real time system

    (b) Main frame system.

17. Describe SJF and RR CPU scheduling algorithms. With your own CPU burst and arrival time.

18. Discuss deadlock prevention and avoidance methods.

19. Describe paging and segmentation memory management schemes.

20. Explain briefly about the way of accessing a file.

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RW–10048 **545203/541203**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Applications

COMPUTER NETWORKS

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define: Internet.

2. What is ATM in Computer Networks?

3. What is CRC?

4. What are Bridges in Computer Networks?

5. Define: Virtual Circuits.

6. What are Static Routing?

7. Define: Congestion.

8. Write the purpose of Network Layer.

9. What is TCP?

10. Define: Multimedia.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short notes on: APRANET.

Or

(b) Discuss about Narrowband and Broadband ISDN.

12. (a) Discuss the design issues of Data Link Layer.

Or

(b) Write short notes on: Wireless LANs.

13. (a) Discuss the differences between Virtual Circuit and Datagram Subnets.

Or

(b) Write about Distance Vector Routing with example.

14. (a) What are the policies that affect Congestion? Explain them.

Or

(b) Discuss about Connectionless internetworking and Tunneling.

15. (a) What are Transport Service Primitives? Explain them.

Or

(b) Write about SNMP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the various Transmission Media used for Computer Networks.

17. Explain about any two Elementary Data Link Protocols.

18. Describe the working of Hierarchical and Shortest Path Routing with suitable examples.

19. Explain about IP Addresses and Internet Control Protocols.

20. Discuss the following :

    (a) The Domain Name System.

    (b) The World Wide Web.

RW–10049 **541204/545204**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Science

COMMUNICATION SKILLS

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the basic techniques required for a fruitful negotiation?

2. What is intra personal communication?

3. Mention any three important barriers to communication.

4. What is known as concord?

5. Mention the types of body language.

6. What is the importance of conducting a mock interview?

7. What is an agenda?

8. What are the main techniques involved for delivering a good presentation?

9. What is meant by a postscript?

10. For what do we prepare minutes?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) How the visual aids help a person to deliver his speech in style?

Or

(b) How can we make a group discussion effective in testing the personality of a candidate?

12. (a) A report should be clear, simple, exact and brief. Why?

Or

(b) What is the difference between formal and informal writing?

13. (a) Why is it that a speaker likes to interact with the audience after his/her talk?

Or

(b) Which do you think will be more effective an extempore or a speech read out from a manuscript?

14. (a) Describe about the most important role played by intonation and stress while delivering a speech.

Or

(b) What are the physical aspects of a speech?

15. (a) Explain the art of conversation using different medias.

Or

(b) Prepare a short speech on the empowerment of women – Is it a reality or a dream?

Part C (3 × 10 = 30)

Answer any **three** of the following.

16. Are negotiations meant only for resolving conflicts or do they have any other purpose in our everyday life?

17. What traits of candidates are evaluated by the panelists during a group discussion? Elaborate your answer by citing suitable examples.

18. What is the concept of a good presentation? What are the characteristics that distinguish it from a written report?

19. Write an essay on how communication provides the network of sustenance essential for the growth and smooth functioning of any organization.

20. Write short notes on :

    (a) Agenda writing

    (b) Downward communication

    (c) Mock interview.

RW–10050 **541205/545205**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

ACCOUNTING AND FINANCIAL MANAGEMENT

(CBCS – 2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are accounting conventions?

2. State the uses of cash flow statement.

3. What is contribution?

4. What is variance analysis?

5. Define master budget.

6. What is the need for budgeting?

7. Define pay-back period.

8. What is risk?

9. Distinguish between networking capital and gross working capital.

10. What is D/P ratio?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Discuss the principles of double entry system.

Or

(b) From the following balance sheets of Arvind Ltd., you are required to prepare a cash flow statement :

Liabilities	2007 Rs.	2008 Rs.	Assets	2007 Rs.	2008 Rs.
Share capital	4,00,000	5,00,000	Land	1,00,000	1,32,000
			Cash	60,000	94,000
Trade creditors	1,40,000	90,000	Debtors	2,40,000	2,30,000
			Stock	1,60,000	1,80,000
Profit & loss a/c	20,000	46,000			
	5,60,000	6,36,000		5,60,000	6,36,000

12. (a) From the following data calculate BEP. Also find the new BEP if selling price is reduced to Rs. 18 per unit.

Budgeted output 1,00,000 units

Fixed expenses Rs. 5,00,000

Variable expenses Rs. 10 per unit

Selling price Rs. 20 per unit.

Or

(b) Discuss the various elements of cost.

13. (a) Explain any two methods of preparing sales budget.

Or

(b) Calculate the earnings per share from the following data :

Net profit before tax Rs. 1,00,000

Taxation 50% of net profit

10% preference share capital (Rs. 10 each) Rs. 1,00,000

Equity share capital (Rs. 10 each) Rs. 1,00,000.

14. (a) Explain the procedure for measurement of risk.

Or

(b) A company is considering investment of Rs. 10,00,000 in a project. The following are the income forecasts after depreciation and tax

I year loss Rs. 1,00,000

II year profit Rs. 3,00,000

III year profit Rs. 4,00,000

IV year profit Rs. 2,00,000

V year profit Rs. 2,00,000

Calculate the ARR.

15. (a) Discuss different types of dividend policy.

Or

(b) The EPS of a company is Rs. 16. The market capitalisation rate applicable to the company is 12.5 percent. Retained earnings can be employed to yield a return of 10%. The company is considering a pay out of 25%, 50% and 75%. Which of these would maximise the wealth of shareholders as per Walter's model?

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss different types of accounting concepts.

17. From the following data, calculate labour variance :

Budgeted labour for completing job X :

8 skilled workers at Rs. 10 per hour for 20 hours

12 unskilled workers at Rs. 8 per hour for 20 hours

Actual labour for completing job X :

12 skilled workers at Rs. 11 per hour for 20 hours

13 unskilled workers at Rs. 7 per hour for 20 hours

18. Discuss the significance of budgetary control in management accounting.

19. KPN Ltd., is considering the purchase of a new machine for Rs. 1,20,000. It has a life of 4 years and an estimated scrap value of Rs. 20,000. The machine will generate an extra revenue of Rs. 4,00,000 p.a. and have additional operating costs of Rs. 3,20,000 p.a. The company's cost of capital is 20% and tax rate 50%. Should the machine be purchased?

20. Discuss the factors affecting working capital.

RW–10026 **541201/545201**

M.C.A. (R) DEGREE EXAMINATION, APRIL 2015

Computer Applications

OBJECT ORIENTED PROGRAMMING AND C++

(Common for M.C.A(R)/M.C.A.(W/E)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the special features of C++ over C?

2. What is the use of cast operator?

3. What is the relationship between a class and an object?

4. Can objects appear as function arguments? Give an example.

5. Define derived class.

6. What is virtual function?

7. Mention the types of templates.

8. What is filling and padding?

9. Write the uses of exception handling.

10. Differentiate between abort( ) and exit ( ) function.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write the benefits of object oriented programming.

Or

(b) How can we design our own manipulators? Give examples.

12. (a) How classes are declared in C++? Explain with an example.

Or

(b) What is a copy constructor? Explain.

13. (a) List the rules used for overloading function names.

Or

(b) Why do we need pure virtual function?

14. (a) Compare function template with macro.

Or

(b) Describe the various file mode options available in C++.

15. (a) How can you use a class object as an exception?

Or

(b) Illustrate memory allocation failure exception with an example.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the basic concepts of object oriented programming.

17. Create a class MAT of size $m \times n$. Define all possible matrix operations for MAT type objects.

18. Explain and illustrate the two types of polymorphism.

19. Explain with an example how to override a function template.

20. Write a program to display error message when C number is divided by zero using exception.